CS 451 – Computational Intelligence
Spring' 2025
Assignment # 3 – Clustering and Meta-heuristic Optimization

Objective: The objective of this assignment is to enable students to gain practical experience in applying fuzzy concepts to clustering methods. Additionally, it promotes critical understanding of meta-heuristic optimization and motivates students to self-learn selected techniques inspired by natural processes.

Question 01 – Fuzziness in Clustering [25 points]

Clustering has wide applications in various fields, including image processing. In this question, you will implement fuzzy-cmeans clustering algorithm for image segmentation which is the task of partitioning image into multiple regions. The implementation will take an image and the number of clusters (C) as input, and it will segment the image into C clusters based on the similarity of pixels, as shown below.

Original Image

Clustered image with K=3

Clustered image with K=5

The resultant image should ideally display each object, or visually similar objects, within the same cluster. In the context of this assignment, **similarity** is determined by the **visual appearance** of pixels, such as their **RGB values.**

a) **[10 Points]** Implement the fuzzy-cmeans algorithm to cluster the given image. The algorithm should take an image and the desired number of clusters (C) as input, and partition the image into C clusters. The algorithm should terminate when there is no significant change in the cluster assignments between iterations. You will experiment with different values of parameters (C and degree of fuzziness) and will analyse and present their results. You should also give your results with different number of clusters.
b) **[05 Points]** Provide a graphical visualization showing the incremental process of cluster formation.
c) **[10 Points]** Compare the performance of fuzzy-cmeans with that of kmeans. You can use some existing implementation of kmeans or have one of your own.

A few sample images are included with the assignment. In addition to submitting your code, you should include the **clustering results** of these images in the **pdf file.**

Question 02 – Meta-heuristic Optimization [10 points]

All of the optimization techniques that we have discussed in the CI course fall in the category of 'Meta-heuristic optimization'. Explain **in your own words**:

- What is meta-heuristic optimization?
- Why is it named so?
- What makes it different from other optimization techniques?

Question 3 – From Nature to Algorithms [15 points]

Select any of the following **swarm intelligence algorithms,** understand it and explain it **in your own words**.

The list of algorithms is given below:

a) Firefly Algorithm (FA)
b) Moth-Flame Optimization (MFO) Algorithm
c) Grey Wolf Optimizer (GWO)
d) Whale Optimization Algorithm (WOA)
e) Artificial Bee Colony (ABC) Algorithm
f) Dragonfly Algorithm (DA)
g) Salp Swarm Algorithm (SSA)
h) Grasshopper Optimization Algorithm (GOA)

Your explanation should cover the following:

1. **Inspiration:** What natural or biological behavior inspires this algorithm? Provide intuition of the algorithm,
2. **Pseudocode:** Provide clear and structured pseudocode for the algorithm.
3. **Exploration vs Exploitation:** How does the algorithm balance exploration (searching new areas) and exploitation (refining known good solutions)?
4. **Diversity Maintenance:** How does the algorithm maintain population diversity?
5. **Knowledge Sharing:** Do agents (particles/individuals) share information? If yes, explain how knowledge is exchanged and influences the search process.
6. **Applicability:** Describe the general type of problem that this algorithm is most suited for.

Submission: You will submit a python notebook for the code and a pdf for analysis and discussion.